

Concepts of Bayesian Data Analysis [G0B74a]

Project 3: Car insurance claims

New Group 27
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Question 1

Write out BUGS code for the Bayesian Negative Binomial regression model for the number of insurance claims. Assume uninformative priors for all the parameters. Explain your code, indicating what is your likelihood, what priors you have chosen, and explain all the parameters.

Before writing the model, we first need to prepare the data. JAGS doesn't understand variables like "District 1" or "<25", so we converted everything to numbers with *as.numeric()*. We stored all the variables: claims, holders, district, age, group, and the number of observations N, in a named list called *jags_data*, which is the input format required for JAGS. Then as JAGS is a separate program from R, it doesn't read R code but it reads text. So we wrote the model in JAGS's with the BUGS language and saved it as a text file. The BUGS is called "*bay_kul_model.txt*", and its content will be explained in the following paragraphs.

Additionally, we computed the mean of $\log(\text{Holders})$. For this model, we subtract this mean from $\log(\text{Holders}[i])$ so that the offset is centered around zero. This is a centering technique that reduces posterior correlations between the intercept and the other coefficients which improves the mixing of the MCMC sampler. The effect of this choice on the diagnostics is discussed in Question 2.

Our BUGS model (see Model 1) has three main components: the likelihood, the linear predictor, and the priors.

```
model {  
  for (i in 1:N) {  
    Claims[i] ~ dnegbin(p[i], r)  
    p[i] <- r / (r + mu[i])  
    log(mu[i]) <- (log(Holders[i]) - log_Holders_mean) + beta0  
      + beta1 * equals(District[i], 2)  
      + beta2 * equals(District[i], 3)  
  }  
}
```

```

+ beta3 * equals(District[i], 4)
+ beta4 * equals(Age[i], 2)
+ beta5 * equals(Age[i], 3)
+ beta6 * equals(Age[i], 4)
+ beta7 * equals(Group[i], 2)
+ beta8 * equals(Group[i], 3)
+ beta9 * equals(Group[i], 4)
}
beta0 ~ dnorm(0, 0.0001)
beta1 ~ dnorm(0, 0.0001)
beta2 ~ dnorm(0, 0.0001)
beta3 ~ dnorm(0, 0.0001)
beta4 ~ dnorm(0, 0.0001)
beta5 ~ dnorm(0, 0.0001)
beta6 ~ dnorm(0, 0.0001)
beta7 ~ dnorm(0, 0.0001)
beta8 ~ dnorm(0, 0.0001)
beta9 ~ dnorm(0, 0.0001)
r ~ dgamma(0.01, 0.01)
}

```

Model 1

For the likelihood, as specified in the assignment, we assume that the number of claims follows a negative binomial distribution. Our code “*Claims[i] ~ dnegbin(p[i], r)*” is the likelihood. It says “the number of claims in row *i* follows a negative binomial distribution.” However, JAGS uses a specific parameterization for this distribution and requires a probability **p**, so we have to convert our expected claim count **mu** into a probability using the formula **p = r / (r + mu)**, which is derived from the relationship **E[Yi] = r(1 - p) / p** given in the assignment instructions.

Once we have the likelihood, we need to define the predictor. We model $\log(\mu)$ rather than μ directly. This is because μ represents a count of claims, which can never be negative. By working on the log scale, we can know that μ will always be positive. In other words, no matter what values the betas take, when we backtransform to our scale (by taking the exponential), the result is always a positive number. We also include $\log(\text{Holders}[i])$ so that the model compares claim rates per policyholder rather than raw claim counts, since groups with more policyholders will naturally have more claims regardless of how risky they are.

In our BUGS code, to capture the effects of our variables, we use the `equals()` function to create dummy variables. For example, `equals(District[i], 2)` simply asks "is this group in District 2?" If yes, it returns 1 and the corresponding beta gets added to the equation. If no, it returns 0 and that beta is ignored. We do this for categories 2, 3, and 4, which gives us 3 dummy variables per predictor. The first category is left out because it serves as the reference level and all the other categories are compared to it.

Finally, we need to specify our priors. We use uninformative priors for all parameters, as requested in the assignment. The 10 regression coefficients (β_0 to β_9) each receive a normal prior with mean 0 and precision 0.0001, meaning a variance of 10,000. This is so wide that the prior has almost no influence on the result, the data will determine the final estimates. We chose a normal distribution for the betas because they can be positive or negative (for example, living in a certain district could increase or decrease the claim rate).

The dispersion parameter r receives a $\text{gamma}(0.01, 0.01)$ prior, which is also uninformative. We cannot use a normal prior for r because r must be strictly positive, a negative dispersion would not make any sense (we cannot have a negative dispersion) and the gamma distribution only produces positive values.

These are our parameters mentioned in our BUGS model:

Beta 0: the intercept is the baseline log claim rate when all covariates are at their reference level (District 1, Age <25, Car group <1L) at the average log number of policyholders
Beta1, beta2, beta3: the effect of being in District 2, 3, or 4 compared to District 1
Beta4, beta5, beta6: the effect of being in Age group 25–29, 30–35, or >35 compared to Age <25
Beta7, beta8, beta9: the effect of being in Car group 1–1.5L, 1.5–2L, or >2L compared to Car group <1L
r: the dispersion parameter which controls how much extra variability there is
$\text{Log}(\mu[i])$: the log of the expected number of claims for group i
$p[i]$: the probability parameter for the negative binomial distribution, computed from $\mu[i]$
$\text{log}(\text{Holders}[i])$: accounts for the different number of policyholders in each group. So the model now compares claim rates per policyholder rather than raw claim counts, since groups with more policyholders will naturally have more claims regardless of how risky they are.

Table 1

Question 2

Run the MCMC algorithm. Describe your choices (number of chains, iterations, burn-in). Assess convergence of the chains using appropriate diagnostics and summarize your conclusions.

Now that we have defined the model in question 1, we need to actually run it and obtain posterior samples. We run the MCMC algorithm using JAGS, which uses a combination of Gibbs sampling and other MCMC methods such as Metropolis.

Before running the algorithm, we need to give it a starting position called “the initial values”. We created 3 chains, meaning we ran the algorithm 3 times from 3 different starting positions, so *n.chains* = 3. If all 3 chains end up exploring the same target

distribution, this shows us that they have converged to the actual posterior. 3 chains is a good number as 2 is the minimum for diagnostics.

For our initial values, we use a random generator function where coefficients are drawn from a normal distribution. If all chains arrive at the same posterior distribution with these starting points, it provides evidence that the model has converged. For r we used `runif(1, 0.5, 5)` instead of `rnorm` because r is a parameter that must be strictly positive, and `rnorm` could generate negative values. For `runif(1, 0.5, 5)`, the lower bound of 0.5 keeps r away from zero, since a value of r at zero would cause numerical problems in its formula.

Next, we need to tell JAGS which parameters to track and save. We told JAGS to monitor all of our parameters: `parameters <- c("beta0", "beta1", ..., "r")`. With everything prepared, we can compile our model. We loaded the model with the previous text file `"bay_kul_model.txt"`, with the data we prepared `"data = jags_data"` and its initial values `"my.inits"` and the number of chains `"n.chains=3"`.

We first tried the following: burning 1000, with 10,000 iterations. In this case, autocorrelation was very high and there were not enough independent draws, in general, diagnostics were not optimal. We then tried burning 1000, 10,000 iterations, with a thinning factor of 3, and we obtained 3,000 samples per chain, giving 9,000 total samples. However, the effective sample size for β_0 was only 987, so our 6,000 samples only carried the information of 987 independent draws, and autocorrelation was still not good even after lag 20 and 30.

Rather than increasing the thinning factor, we applied centering which is an acceleration technique that attacks the root cause of the autocorrelation. We subtracted the mean of `log(Holders)` from `log(Holders[i])` in the model. By subtracting the mean of `log(Holders)` in the model, the offset now is around zero, and β_0 can move independently from the other parameters. This reduces posterior correlations between the intercept and the other coefficients, allowing the sampler to mix faster.

We also increased to a burn-in of 5000, 50,000 iterations with no thinning factor, giving 45,000 samples per chain and 135,000 total. After this change, the effective sample size for β_0 improved to 4,361, confirming that the chains now produce enough independent draws for posterior estimation. We chose a burn-in of 5,000 because while the regression coefficients converged quickly, the parameter r required a longer period to reach its posterior region (has a mean of 188) from its initial starting range (0.5–1).

With these samples, we verified that the MCMC algorithm and its collected samples truly represent the posterior distribution. To do this, we use several diagnostics. So we check convergence from multiple angles: trace plots, the gelman rubin diagnostic, effective sample sizes, and autocorrelation plots.

The trace plots show that the chains for β_0 through β_9 seem to have converged, since the three chains overlap and move around stable values. This means we can trust the posterior estimates. For r the trace plot (Figure I) does not look as good. The chains stay near low values for a while and then suddenly jump to high values, which suggests the sampling has trouble exploring the posterior. We can also see that the density plot of r (Figure II) is skewed to the right with a long tail.

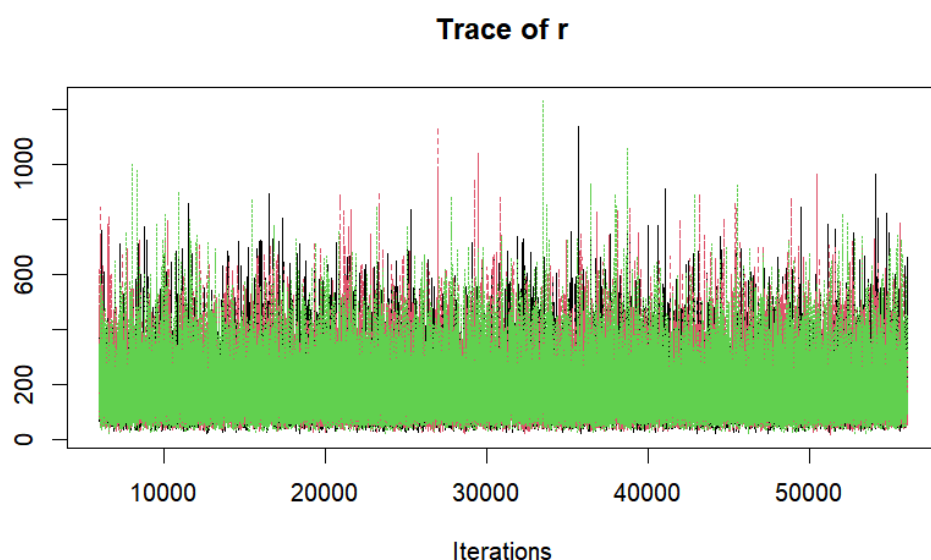


Figure I

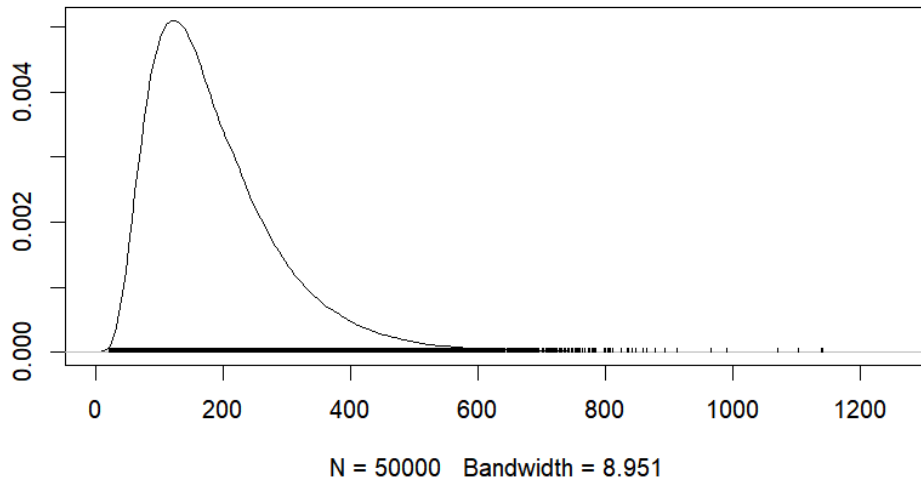


Figure II

Also, the posterior densities of all beta parameters are approximately symmetric, it is just the posterior of r which is right-skewed with a long tail. (Figure II).

The Gelman-Rubin diagnostic compares the variance within each chain to the variance between chains. If the chains have converged to the same distribution, the two variances should be similar, giving a shrink factor close to 1. In our case, all parameters have a point estimate of 1 and upper confidence intervals of 1 as well, which indicates convergence. The Gelman-Rubin plots (Figure III and Figure IV) confirm this, as it starts higher in the early iterations but drops fast and stabilizes near 1 for all parameters. The multivariate potential scale reduction factor is also 1, which means that the chains have converged when considering all parameters together.

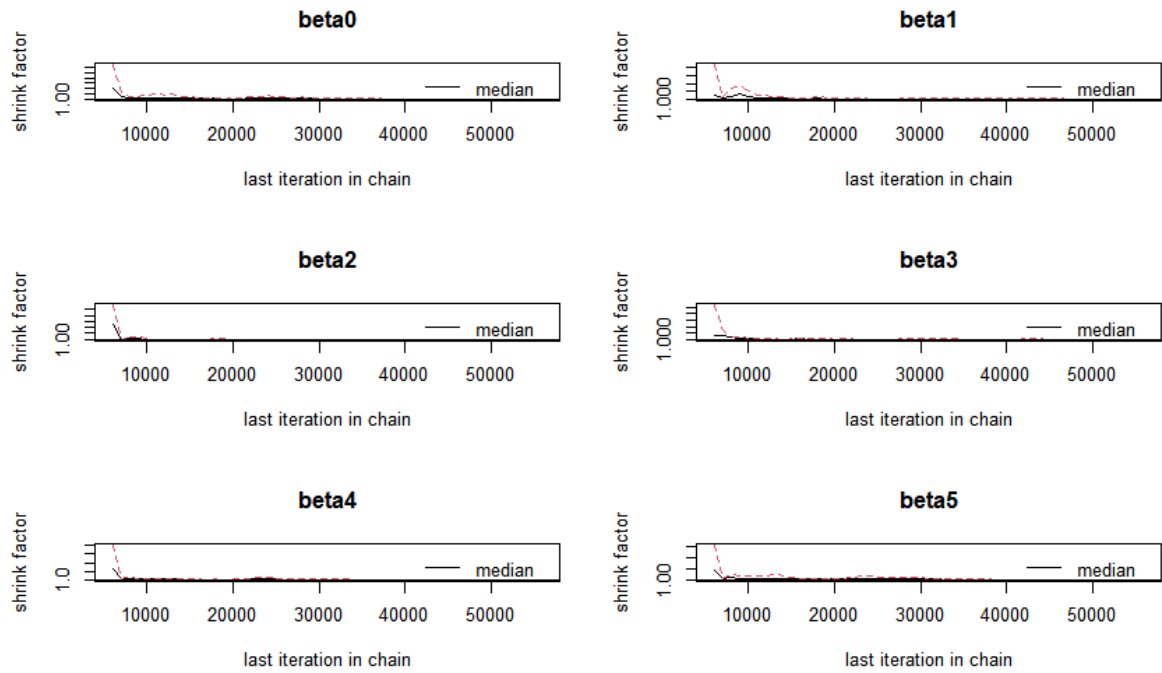


Figure III

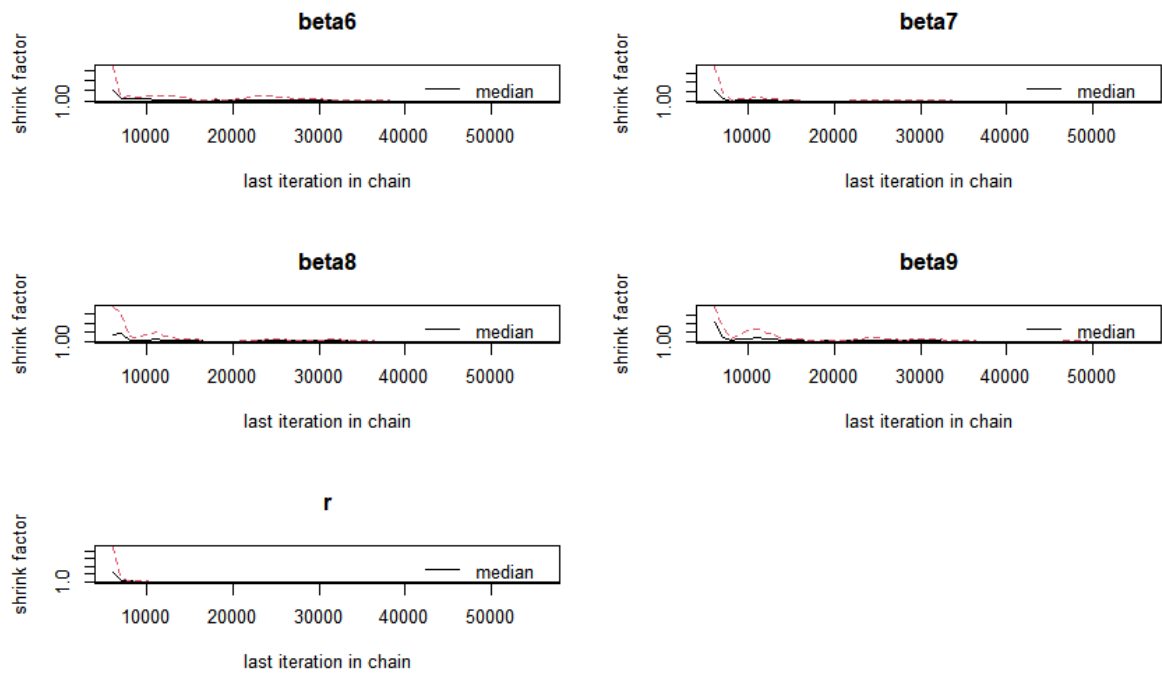


Figure IV

The effective sample sizes (Figure V) for all parameters are ranging from 4361 for β_0 to 36741 for r . This means that, despite the autocorrelation that is typical in MCMC

samples, the chains produced enough independent draws for each parameter to get stable posteriors.

beta0	beta1	beta2	beta3	beta4	beta5	beta6	beta7	beta8
4361.927	32362.330	37203.903	42401.772	7876.259	7506.145	6052.363	14330.144	15981.040
beta9	r							
22755.093	36741.190							

Figure V

In the previously-mentioned first run with 10,000 iterations, burn in of 1000, and no thinning (=1), the autocorrelation plots show high autocorrelation that decays slowly, see Figure VI. This means that consecutive samples are highly correlated with each other. The final autocorrelation plots from our last try with the centering and the 50000 iterations do not show a big improvement but eventually the autocorrelation drops to near zero for all parameters, see Figure VII. However, the effective sample sizes are higher and the posterior estimates are reliable.

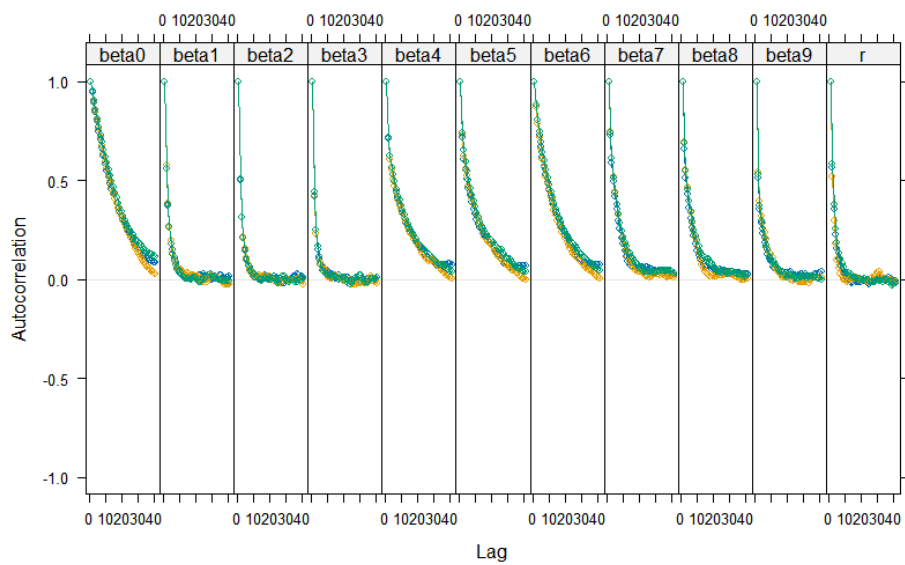


Figure VI

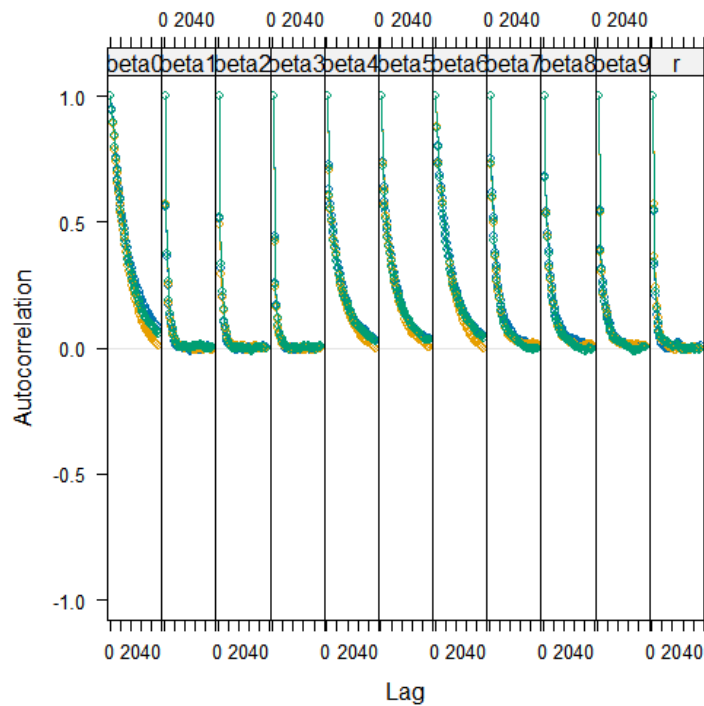


Figure VII

To summarize, we ran 3 chains starting from different random initial values to check whether they all converge to the same area. We finally used a burn-in of 5,000 iterations to discard the early samples, and then collected 50,000 iterations per chain with no thinning factor. We initially tried 10,000 iterations with low thinning, but the autocorrelation plots showed that consecutive samples were too correlated and the sample sizes were low. After all of our adjustment, all diagnostics confirm convergence: the trace plots show stable and overlapping chains, except for r . The Gelman-Rubin shrink factors are all 1 with upper bounds of 1, and with a multivariate factor of 1. The effective sample sizes range from 4,361 to 36,741, and the autocorrelation drops to near zero eventually.

The posterior of r is right-skewed with a long tail, which is expected for a positive dispersion parameter in a negative binomial model. The beta coefficients are unconstrained coefficients on the real line, so their posteriors are symmetric and normal-shaped. Overall, the sampler has converged and we can trust the posterior estimates.

Question 3

Give (point and interval) posterior summaries for all the parameters in the model. Interpret the effects of age, car group and district.

Now that we have obtained our posterior samples in question 2, we can summarise the posterior distributions of all parameters. The posterior summary includes a point estimate (posterior mean) and an interval estimate, the 95% credible interval, given by the 2.5% and 97.5% quantiles. The 95% credible interval can be interpreted as “there is a 95% posterior probability that the true parameter value lies within this range, given the data”. The results are presented in Table II.

All effects are interpreted holding the other variables constant at their reference levels, that is District 1, age under 25, and engine size under 1 litre.

Interpreting district effects ($\beta_1, \beta_2, \beta_3$), β_1 and β_2 have small positive means of 0.0356 and 0.0427 respectively, but both credible intervals include zero, meaning we have no strong evidence that these districts differ from District 1 in their claim rate. District 4 has a posterior mean of 0.2319 with a credible interval above 0, giving us evidence that it is associated with more claims than the reference District 1.

$\beta_4, \beta_5, \beta_6$ are all negative and all exclude 0 from their credible interval. This means that every older age group files fewer claims than the under-25 reference group, and the effect strengthens with age. The 25–29 group ($\beta_4 = -0.1855$) shows a modest reduction, the 30–35 group ($\beta_5 = -0.3373$) a larger one, and the over-35 group ($\beta_6 = -0.5285$) the largest reduction of all.

For car groups ($\beta_7, \beta_8, \beta_9$), all demonstrate the opposite effect, with larger engine size being associated with higher claim rates (all have positive means). The effect increases with group size, from $\beta_7 = 0.1509$ for the 1–1.5 litre group up to $\beta_9 = 0.5580$ for the over 2 litre group. The credible interval does not include 0 for any engine type, meaning there is evidence that these groups differ from the reference group of cars with engines less than 1L.

Finally, the dispersion parameter r has a posterior mean of 186.9834 with a very wide credible interval from 56.1643 to 451.0768. In the Negative Binomial distribution, larger values of r correspond to less overdispersion, approaching the Poisson distribution. This suggests that the included covariates effectively capture the underlying structure of the data, leaving relatively little unexplained variability in the claim counts.

Parameter	Mean	SD	2.5%	Median	97.5%
beta0	3.0750	0.0873	2.9016	3.0757	3.2437
beta1	0.0356	0.0587	-0.0786	0.0354	0.1522
beta2	0.0427	0.0657	-0.0865	0.0427	0.1718
beta3	0.2319	0.0745	0.0834	0.2323	0.3765
beta4	-0.1855	0.0920	-0.3646	-0.1856	-0.0049
beta5	-0.3373	0.0906	-0.5130	-0.3376	-0.1585
beta6	-0.5285	0.0805	-0.6848	-0.5288	-0.3686
beta7	0.1509	0.0644	0.0242	0.1511	0.2759
beta8	0.3944	0.0681	0.2609	0.3943	0.5284
beta9	0.5580	0.0834	0.3937	0.5581	0.7211
r	186.9834	104.3797	56.1643	163.2595	451.0768

Table II. Posterior Summaries of Regression Coefficients and Dispersion Parameter

Question 4

The regression coefficients can be interpreted through rate ratios:

$$RR_k = \exp(\beta_k).$$

Compute posterior summaries for the rate ratios and interpret the results.

Using regression coefficients obtained in question 3 we can further interpret the effects multiplicatively through rate ratios ($\exp(\beta_k)$). The posterior summaries for the rate ratios are given in the Table III below.

Parameter	Mean	Median	2.5%	97.5%
RR0 (Intercept)	21.7319	21.6649	18.2026	25.6296
RR1 (District 2)	1.0380	1.0361	0.9244	1.1644
RR2 (District 3)	1.0459	1.0437	0.9171	1.1874
RR3 (District 4)	1.2644	1.2615	1.0869	1.4571
RR4 (Age 25–29)	0.8342	0.8306	0.6945	0.9951
RR5 (Age 30–35)	0.7167	0.7135	0.5987	0.8534
RR6 (Age >35)	0.5914	0.5893	0.5042	0.6917
RR7 (Group 1–1.5L)	1.1653	1.1631	1.0244	1.3177
RR8 (Group 1.5–2L)	1.4869	1.4834	1.2980	1.6962
RR9 (Group >2L)	1.7532	1.7473	1.4825	2.0567

Table III. Posterior Summary of Rate Ratios ($\exp(\beta)$)

A rate ratio greater than 1 indicates that a group has a higher claim rate than the reference category, while a rate ratio below 1 indicates a lower claim rate. If the credible interval includes 1, the effect is not considered credible, if it excludes 1, there is evidence of a credible effect.

These results are consistent with those obtained in Question 3. We observe that the effects for Districts 2 and 3 are not credible, while all remaining coefficients indicate credible differences. Simultaneously, larger engine sizes, younger age, and residing in District 4 are associated with more claims.

Looking at the district effects, Districts 2 and 3 have rate ratios of 1.0380 and 1.0459, suggesting claim rates roughly 4% higher than District 1 with no credible difference.

District 4 has a rate ratio of 1.2644, meaning we are confident that it generates approximately 26% more claims than District 1.

For age, the 25–29 group files around 17% fewer claims (RR = 0.8342), the 30–35 group around 28% fewer (RR = 0.7167), and the over-35 group around 41% fewer (RR = 0.5914).

For car group, the 1-1.5 litre group has about 17% more claims than the reference (RR = 1.1653), the 1.5-2 litre group about 49% more (RR = 1.4869), and the over-2 litre group about 75% more (RR = 1.7532). This is a strong and monotonically increasing effect, suggesting that engine size is one of the more influential predictors in the model.

Question 5

Predict the number of insurance claims for each age group in District 1, Car Group 1, and 100 policyholders. Give summary measures and plots of the posterior predictive distributions. Interpret the results.

To predict the number of insurance claims per age group, it is necessary to derive the form of the expected value. The relationships $E[Y_i] = \mu_i$ and $\log(\mu_i) = \log(E_i) + \eta_i$ are known and the latter can be inserted into the former one to obtain:

$$E[Y_i] = \exp(\log(E_i) + \eta_i) = \exp(\log(E_i) + \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_9 x_{9i}).$$

Since we are looking at district 1 and car group 1, we know that $x_{ji}=0$ for j in $\{1, 2, 3, 7, 8, 9\}$ and all i . The district and car group are reflected by the intercept β_0 . Additionally, we need to subtract the mean of the logarithm of the number of policyholders, as the model is centered around this variable. This means that instead of $\log(E_i)$ we consider $\log(E_i) - \text{mean}(\log(E_i))$. We insert $E_i=100$. For the four age groups we obtain the following, by setting the corresponding x_{ji} to 0 or 1:

$$<25: E[Y_i] = \exp(\log(100) - \text{mean}(\log(E)) + \beta_0)$$

$$25-29: E[Y_i] = \exp(\log(100) - \text{mean}(\log(E)) + \beta_0 + \beta_4)$$

$$30-35: E[Y_i] = \exp(\log(100) - \text{mean}(\log(E)) + \beta_0 + \beta_5)$$

$$>35: E[Y_i] = \exp(\log(100) - \text{mean}(\log(E)) + \beta_0 + \beta_6).$$

By rounding the mean of the transformed samples, we obtain the predicted number of claims per age group. For the age group <25, the predicted number of claims is 16 (16.115), for 25-29 year olds, it is 13 (13.376) claims, for 30-35 it is 11 (11.492) claims and for over 35 year olds it is 9 (9.483). Overall, these results show that the expected number of claims decreases with age, when we are looking at district 1, car group 1 and 100 policyholders.

Next, we sample from the posterior predictive distribution to obtain summary measures and density plots. We use the parametrization of *rnbinom* with the dispersion parameter r as size and the above calculated claims as the mean μ . The parameter r is extracted from the matrix containing the MCMC samples. The summary measures are given in Table IV.

Age group	Mean	Median	Variance	2.5%	97.5%
<25 (group 1)	16.111	16	19.900	8	26
25-29 (group 2)	13.386	13	15.710	6	22
30-35 (group 3)	11.510	11	13.249	5	19
>35 (group 4)	9.472	9	10.416	4	16

Table IV: Summary measures of the posterior predictive distributions

The summary measures of the PPD show us that for future predictions, with increasing age, a decrease in claims can be expected. However, as is reflected by the big variances and overlapping 95% credible intervals, there is much insecurity about predicting future observations and no clear differentiation between the age groups is possible. The density plots of the PPD are shown in Figure VIII. We see that the higher the age is, the narrower the distribution is and the more it is shifted to the left.

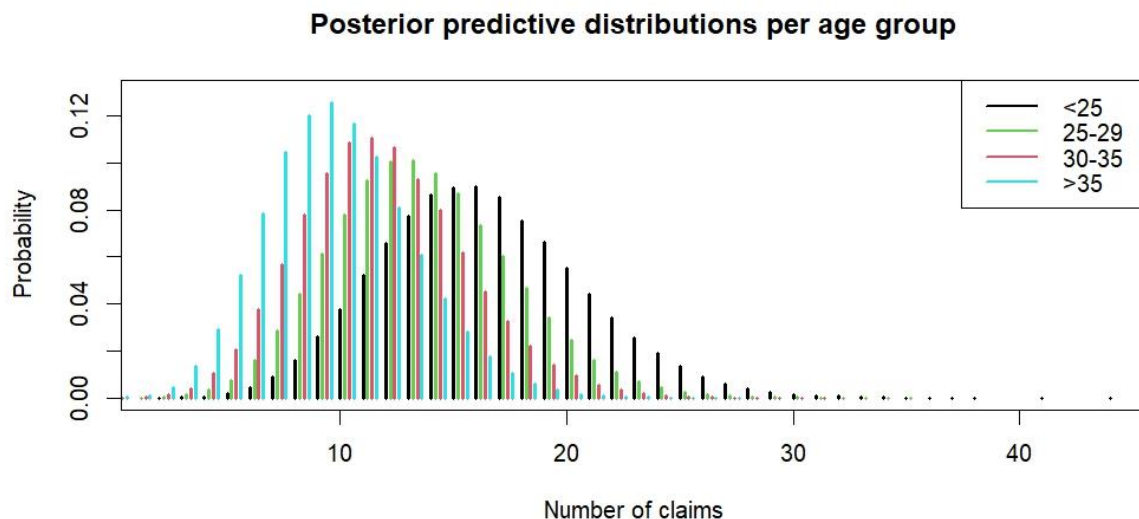


Figure VIII: Density plots of the posterior predictive distributions

Question 6

Estimate the posterior probability that the claim rate for policyholders of age 25-29 is higher than the claim rate for policyholders of age 30-35: $P(\mu_{25-29} > \mu_{30-35} | \text{data})$. Explain how this probability is obtained.

To estimate the posterior probability that the claim rate for policyholders of age 25-29 is higher than the claim rate for policyholders of age 30-35, we can use the estimated posterior claims from task 5. For every posterior value, we can compare whether the expected number of claims is higher in the age group 25-29 and sum up all those cases, dividing them by the total chain length, which is obtained in R by the command `"sum(claims_25_29 > claims_30_35)/length(claims_25_29)"`. The estimated probability is 97.46%, meaning that almost always, the number of claims is higher in the age group 25-29 than in group 30-35.

Question 7

Compute the posterior distribution of claim rate per 100 policyholders of the youngest age group with a car of group 1 per district. Give posterior summaries, and visualize the posterior distributions using a caterpillar plot. Interpret the result.

As we are talking about claim rates here, we must first consider the model again. Given the centering choice made in Question 1, the model now looks as follows:

$$\log(\mu_i) = \log(E_i) - \text{avg}(\log(E_i)) + \eta_i$$

Where μ_i is the expected number of claims, E_i the policyholders (exposure) and η_i the effects of the covariates. In order to consider claims rate, we rearrange the model to obtain μ_i / E_i :

$$\mu_i = \exp(\log(E_i) - \text{avg}(\log(E_i)) + \eta_i)$$

$$\mu_i = E_i * \exp(- \text{avg}(\log(E_i)) + \eta_i)$$

$$\mu_i / E_i = \exp(- \text{avg}(\log(E_i)) + \eta_i)$$

This is then the claim rate per policyholder, but we're interested in the claims rate per 100 policyholders, so we multiply both sides by 100:

$$100 * \mu_i / E_i = 100 * \exp(- \text{avg}(\log(E_i)) + \eta_i)$$

We implement this in R by creating an object per district, so that through η_i we may observe the claim rate per 100 policyholders per district, while keeping age and car group constant at their baseline.

For example, the claim rate per 100 policyholders for district 1 of the youngest age group and car group 1 is obtained in R through

```
rate_d1 <- 100 * exp(-log_Holders_mean + results_matrix[, "beta0"]),
```

where *results_matrix* is a matrix object of the posterior MCMC samples. Such a claim rate object is made for each of the four districts, where subsequent districts are constructed through adding their respective *results_matrix[, "betaX"]* to the exponent denoted above.

The theoretical formulation of these objects are as follows:

District 1: $100 * \exp(- \text{avg}(\log(E_i)) + \beta_0)$

District 2: $100 * \exp(- \text{avg}(\log(E_i)) + \beta_0 + \beta_1)$

District 3: $100 * \exp(- \text{avg}(\log(E_i)) + \beta_0 + \beta_2)$

District 4: $100 * \exp(- \text{avg}(\log(E_i)) + \beta_0 + \beta_3)$

And so by using the respective MCMC samples for the beta parameters (as shown in the R code above) we obtain the summary statistics found in Table V.

District group	Mean	Median	Variance	2.5%	97.5%
District 1	16.115	16.065	1.976	13.498	19.005
District 2	16.707	16.639	2.416	13.854	19.942
District 3	16.837	16.758	2.822	13.759	20.343
District 4	20.356	20.257	4.640	16.417	24.836

Table V: Summary statistics of the claim rate per 100 policyholders, for age group <25 and car group 1.

Since we hold age and car group constant at their baseline category, we are in effect comparing the underlying claim intensity across districts for the same type of driver. For district 1, we see that the model predicts on average about 16 claims per 100 policyholders for those that are younger than 25 years old and are in car group 1 (meaning with engine size <1l). The credible interval further suggests there to be a 95% posterior probability that the true claim rate lies between 13.5 (≈ 14) and 19 claims per 100 policyholders.

For Districts 2 and 3, the posterior mean claim rates are 16.7 and 16.8 claims per 100 policyholders. These are only slightly higher than District 1, and their credible intervals overlap strongly with that of District 1. This is consistent with the results from Question 4, where there was no strong evidence that Districts 2 and 3 differ from District 1.

District 4 on the other hand has an average claim rate of 20.4, indicating that young policyholders in car group 1 are expected to generate roughly 20 claims per 100 policyholders, compared to the 16 for those of district 1. This is substantially higher, and aligns with the rate ratio of 1.265 ($\approx 20.356/16.115$) obtained in Question 4.

Next, we visualize these results in a Caterpillar plot through the R package `ggmcmc`. Since the claims are computed afterwards instead of directly through the sampling in JAGS, we first put our claim rates for each district in a single matrix, which we convert to an MCMC object through `as.mcmc()`, and convert to `ggmcmc` format through `ggs()`. The resulting `ggs_caterpillar()` is shown in Figure IX.

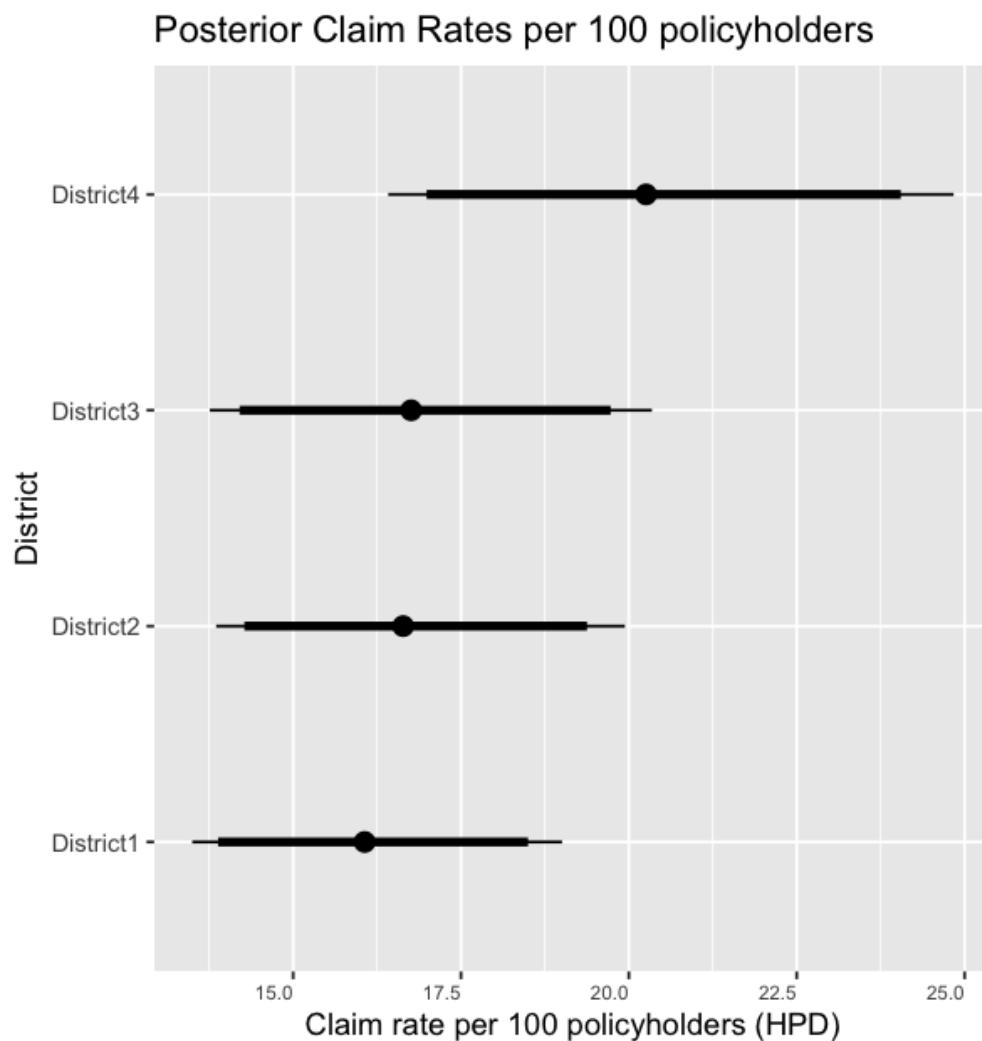


Figure IX: Caterpillar plot of the posterior claim rates per 100 policyholders, per district, held constant for age group <25 and car group 1.

The caterpillar plot visualises for each district the posterior distribution of the claim rates per 100 policyholders. Per district, we then visualise the median and 95% highest posterior density interval (thin line), meaning the narrowest range containing the highest density 95% of posterior values. The wider this interval, the greater the posterior uncertainty about the true claim rate. This is in contrast to what we summarised in table

V, where we saw the equal tail credible intervals. However, since the mean and median values observed indicate approximate symmetry, we note that the two type of credible intervals are practically equivalent.

For district 1, 2 and 3, we see very similar posterior claim rates, again suggesting that no strong evidence exists that these districts differ substantially in their underlying claim intensity for young policyholders in car group 1. District 4 on the other hand stands out clearly as having a posterior claim rate that is noticeably higher than that of the others. We simultaneously note that the HPD interval of district 4 is much wider than that of the other three, indicating more uncertainty about the true posterior claim rate for that district, within this category (young people and small car engines).

From the caterpillar plot, we also note that the lower bound of district 4 is around the mean of district 3, which further visually confirms a meaningful shift of its posterior distribution of claim rates to the right. In other words, most plausible values of district 4 are larger than the central values of the other districts.

Question 8

Investigate whether the parameter estimates are robust to the prior choice for the dispersion parameter.

In order to investigate whether the posterior inference is robust to the prior specification of the dispersion parameter r , we perform a prior sensitivity analysis.

Up to this point, the model used a vague/uninformative Gamma prior,

$$r \sim \text{Gamma}(0.01, 0.01),$$

which places probability mass over a very wide range of positive values. To then assess sensitivity to this choice, we compare the resulting posterior distributions to those obtained under alternative prior specifications.

Since the dispersion parameter must remain strictly positive (variance cannot be negative), all alternative priors are chosen on the positive real line. We therefore investigate both (1) sensitivity to the hyperparameter choice within the Gamma family, as well as (2) sensitivity to the choice of prior family itself through a log-normal specification.

For the Gamma family, we compare the initial vague prior to the more informative prior

$$r \sim \text{Gamma}(1,1)$$

Which concentrates more probability mass around moderate values of r .

Through this log-normal specification, we similarly compare both a vague prior to a more informative one. We do so by modeling the logarithm of the dispersion parameter though

$$\text{Log}(r) \sim N(0,100)$$

And

$$\text{Log}(r) \sim N(0,1)$$

Where the first represents a highly diffuse prior on the log scale and the second a regularized alternative. We therefore place a normal prior on the logarithm of r , and then in the BUGS model exponentiate that variable to recover r . The exponentiating guarantees the dispersion parameter to be constrained to $r > 0$.

As such we explicitly examine whether the posterior conclusions depend on either the strength or the family of the prior distribution parameter. The BUGS code of the various scenarios we tested are summarized by table VI.

Prior Family	Vague	More informative
Gamma	$r \sim \text{dgamma}(0.01, 0.01)$	$r \sim \text{dgamma}(1, 1)$
Log-normal	$\log_r \sim \text{dnorm}(0, 0.01)$ $r <- \exp(\log_r)$	$\log_r \sim \text{dnorm}(0, 1)$ $r <- \exp(\log_r)$

Table VI: Summary of BUGS code for Sensitivity analysis, comparing vague and more informative priors of the gamma and log-normal family.

The BUGS code of the model as a whole, aside of these changes to the dispersion parameter r , remains identical to that of Question 1, and we refer to that section of this document for the full specification.

For the log-normal specification in BUGS/JAGS, we further note that the symbol $\sim$ denotes that a variable follows a probability distribution, reflecting the stochastic uncertainty inherent in Bayesian sampling. In contrast, the symbol $<-$ denotes a deterministic relationship. Hence, in $\log_r \sim \text{dnorm}(0, 0.01)$, the variable $\log_r$ is sampled from a Normal distribution, whereas in $r <- \exp(\log_r)$, the parameter r is obtained deterministically as the exponential transformation of the already sampled $\log_r$.

For the log_normal specification, we further have to specify the initial values list as $\log_r = \log(\text{unif}(1, 0.5, 5))$ instead of having random uniform samples for r itself. This again results from the fact that r is specified deterministically in our BUGS code. We thus take the logarithm of the random unified sample, so that the subsequent exponential transformation gives values in line with the initial values of the other models.

The resulting summary measures of our sensitive analysis are denoted in table VII.

Parameters	(1) Gamma Vague	(2) Gamma informative	(3) Log-normal Vague	(4) Log-normal informative
β_0 (Intercept)	3.0750	3.0588	0.6800	3.0729
β_1 (District 2)	0.0356	0.0783	-11.490	0.0481
β_2 (District 3)	0.0427	0.0491	-4.975	0.0455
β_3 (District 4)	0.2319	0.2156	-10.360	0.2296
β_4 (Age 25-29)	-0.1855	-0.1884	-21.390	-0.1875
β_5 (Age 30 – 35)	-0.3373	-0.3281	-5.311	-0.3363
β_6 (Age > 35)	-0.5285	-0.5068	-10.630	-0.5252
β_7 (Group 1-1.5L)	0.1509	0.1416	-5.425	0.1442
β_8 (Group 1.5 - 2L)	0.3944	0.4046	-6.538	0.3958
β_9 (Group > 2L)	0.5580	0.5571	-4.101	0.5556
r	186.98	12.70	$1.325 * 10^{24}$	70.10

Table VII: Posterior mean estimates under alternative prior specifications for the dispersion parameter r .

- Discussion problematic prior

We first note that the highly diffuse log-normal prior specification for the dispersion parameter results in numerically unstable posterior estimates. This can be explained by considering the implied prior distribution. The prior is specified on the logarithmic scale, where a precision of 0.01 corresponds to a variance of 100. Since the dispersion parameter is obtained through exponentiation, this induces astronomically large plausible values for r on the original scale (e^{100}). Consequently, the resulting prior becomes extremely diffuse and unstable numerically.

This illustrates that excessively broad priors are not necessarily noninformative in practice. In this case, exponentiation amplifies uncertainty to such an extent that the resulting posterior inference becomes unstable, rendering this prior specification unsuitable for the model. The corresponding estimates are therefore excluded from further substantive discussion.

Nevertheless, this finding remains informative for the robustness analysis, as it demonstrates that the parameter estimates are sensitive to extremely diffuse prior specifications for the dispersion parameter.

- Discussion posterior mean

Comparing the vague Gamma prior (1), informative Gamma prior (2), and informative log-normal prior (4), we observe that the regression coefficients remain highly stable across specifications. In particular, both the sign and magnitude of the district, age, and car-group effects are very similar. This indicates that the posterior inference regarding the covariate effects is robust to reasonable changes in the prior distribution of r . In other words, regardless of how we model the extra dispersion, the data still supports roughly the same district, age and car group effects on the expected claim rate.

On the other hand, the posterior mean of the dispersion parameter r changes substantially across prior choices. Under the vague Gamma prior, the posterior mean is approximately 187, while the more informative Gamma prior shrinks this estimate toward 13. The informative log-normal prior produces an intermediate value around 70. We recall that r controls the amount of overdispersion, where a large r brings variance

close to Poisson variance (variance equals mean) and indicates less overdispersion, and a small r implies a much larger variance and hence overdispersion.

From these results we conclude that the data is not very informative about the exact amount of overdispersion, meaning that it has a lot of uncertainty about how much remaining unexplained heterogeneity exists around the mean of the expected claim rates. Nevertheless, the regression coefficients themselves remain comparatively stable, indicating that the covariates robustly explain the systematic variation in the expected claim rates.

- Discussion Credible Intervals and Rate Ratios

We next consider the credible intervals and rate ratios. We here exclude the vague log normal prior, and focus on the other three specifications. Results are shown in table VIII.

Parameter	(1) Gamma Vague	(2) Gamma informative	(4) Log-normal informative
RR0 (Intercept)	21.73 [18.20;25.63]	21.53 [16.04;28.50]	21.71 [17.78;26.11]
RR1 (District 2)	1.04 [0.92;1.16]	1.09 [0.86;1.37]	1.05 [0.91;1.21]
RR2 (District 3)	1.05 [0.92;1.19]	1.06 [0.82;1.35]	1.05 [0.89;1.22]
RR3 (District 4)	1.26 [1.09;1.46]	1.25 [0.95;1.61]	1.26 [1.06;1.49]
RR4 (Age 25–29)	0.83 [0.70;1.00]	0.84 [0.62;1.10]	0.83 [0.68;1.02]
RR5 (Age 30–35)	0.72 [0.60;0.85]	0.73 [0.54;0.96]	0.72 [0.59;0.87]
RR6 (Age >35)	0.59 [0.50;0.69]	0.61 [0.46;0.79]	0.59 [0.50;0.71]
RR7 (Group 1–1.5L)	1.17 [1.02;1.32]	1.16 [0.90;1.47]	1.16 [0.99;1.35]
RR8 (Group 1.5–2L)	1.49 [1.30;1.70]	1.51 [1.17;1.93]	1.49 [1.26;1.75]
RR9 (Group >2L)	1.75 [1.48;2.06]	1.76 [1.33;2.30]	1.75 [1.44;2.10]

Table VIII: Posterior Rate Ratio estimates ($\exp(\beta_k)$) and RR 95% equal-tail Credible Intervals under alternative prior specifications for the dispersion parameter r .

These results show that both the posterior RR estimates and their 95% equal-tail credible intervals remain highly similar across the three (stable) prior specifications. This indicates that the substantive interpretation of the regression effects is robust to reasonable changes in the prior specification of the dispersion parameter r .

We can further draw a direct link between the magnitude of r and the width of the credible intervals. Smaller values of the dispersion parameter imply greater overdispersion, meaning that the model allows more residual variability around the expected claim rates. This additional unexplained variability turns into greater posterior uncertainty for the regression coefficients, resulting in wider credible intervals. This is observed directly in Table VIII. Under the informative Gamma prior (2), where the posterior mean of r is approximately 13, the credible intervals are noticeably wider than under the vague Gamma prior (1), where $r \approx 187$. The informative log-normal prior (4), which produces an intermediate value of $r \approx 70$, correspondingly yields intermediate interval widths.

In terms of the substantive interpretation of the estimates, we observe that under the informative Gamma prior (2), several weaker covariate effects are no longer credibly different from the reference category. In particular, the 95% equal-tail credible intervals for District 4 and Car Group 2 (1–1.5L) now include 1, indicating greater posterior uncertainty regarding whether these groups truly differ from their respective baseline categories.

Under the informative log-normal prior (4), the effect of Car Group 1–1.5L also becomes borderline, with the lower bound of the 95% equal-tail credible interval lying very close to 1 (0.99). Nevertheless, the posterior mean remains highly similar to the other specifications, suggesting that the practical effect size itself remains stable, while the evidence supporting a credibly positive effect becomes slightly weaker due to increased posterior uncertainty.